Sprint Review

< Exploratory Milestone>

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Progress

💡 **What did you accomplish?**

- **User Research & Requirement Engineering:** Developed two primary **Personas**: Leon, the CS Freshman, and Martha, the External Visitor
- **Competitive Analysis:** Performed a feature breakdown of **MazeMap, Pointr, and Google Maps**. Identified the "Unique Selling Point" (USP): Integration of HS-Aalen's **StarPlan (iCal)** directly into an interactive indoor map.
- **Technology Stack Finalized:** Chose **Flutter** for cross-platform development, Python 3.12, PostgreSQL 16, FastAPI(OpenAPI/Swagger), Docker, Nginx
- **Information Architecture:** Completed **Wireframes** for core screens (Dashboard, Map, Settings) and defined **3 Key User Scenarios**.

Next Sprint Goal

💡 **What's planned for next iteration? What is left in total to still do on product backlog (what we agreed on as features)?**

- **Data Modeling:** Set up the Datamodel for Buildings, Rooms, and Events.
- **iCal Parser Implementation:** Create the Scraper for the iCal Timetables.

Critical Aspects

💡 **What Impediments/Risks do you as a team see?**

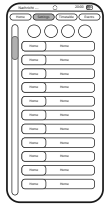
- **Data Quality (Risk):** The accuracy of the app depends on the room naming consistency in iCal + Overview Maps for the Campus.
- **Indoor Mapping Complexity (Impediment):** Converting static PDF floor plans into useable form.
- **iCal Scraper:** Timetable import + scraper

Wireframes

Start

Timetable / Events

Settings



Map / Streetview

Personas

Persona 1: The Anxious Freshman

- **Name:** Leon (23)
- **Archetype:** The "Late-Entry" Freshman (perhaps he did an apprenticeship before starting his degree).
- **Quote:** *"I spend more time looking for the room than actually sitting in it. The university website is a maze I can't solve."*
- **Major:** Computer Science / Electrical Engineering.
- **Tech-savviness: 3/5**. He's competent with apps but gets frustrated by "Academic Software" that feels like it's from 2005.

Goals & Needs ("Why")

- **Finding "The Hidden":** He needs to find the cafeteria (Mensa) and specific labs (like the "Embedded Systems Lab") which are often tucked away in basement levels.

Pain Points & Frustrations ("Ouch")

- **Website Failure:** The new HS-Aalen website is optimized for desktop/marketing, not for a student standing in the rain trying to find a room on a small screen.
- **Naming Logic:** Room codes like "G2 0.21" mean nothing to him. He doesn't know if "G2" is a building or a floor.
- **Information Fragmenting:** His schedule is in one system (LSF/StarPlan), but the map is a PDF somewhere else.

Technical Context ("Environment")

- **Device:** Google Pixel 6a (Android 14).
- **Context:** One hand is usually holding a coffee or a bag; he needs **one-handed navigation** while walking outside.

The "CampusNow" Hook

- **The "Navigation":** Find the best way around the Campus, never miss a Course again

Persona 2: The Concerned Visitor

- **Name:** Martha (52)
- **Archetype:** The "Campus Guest" (Parent/External Visitor)
- **Quote:** *"I just want to meet my son for lunch, but every building looks like a grey concrete block to me."*
- **Relationship:** Mother of a student (or a guest at a public university event).
- **Tech-savviness: 2/5**. She uses WhatsApp and Google Maps for driving, but she finds complex, multi-layered apps overwhelming.

Goals & Needs ("Why")

- **Location Sharing:** She only needs the location ("Meet me at the entrance of Building G 2") and see exactly where that is on a visual map.
- **Point-of-Interest (POI) Search:** She needs to find "Public" spots: restroom, or the cafeteria.
- **Visual Cues:** She needs to see what the building looks like *before* she gets there so she knows she's at the right place.

Pain Points & Frustrations ("Ouch")

- **"Institutional" Language:** She doesn't understand university acronyms like "G 2 0.21" or "Mensa Burren"
- **The "Grey Wall" Effect:** Campus buildings often lack clear, large street numbers. She feels awkward wandering into the wrong building or Campus (Burren vs Main Campus).
- **No Access:** She doesn't have a student login, so if the map is behind a "Student Portal," she can't use it. Also the new HS-Aalen Website is to confusing for here.

Technical Context ("Environment")

- **Device:** iPhone 13.
- **Context:** She is likely walking slowly, looking up at buildings, and then back down at her phone. She needs **large icons** and **clear labels**.

The "CampusNow" Hook

- **"Photo + Location:** The app allows students to send only a location to outsiders. When Martha opens the app and searches for the Room / Building she sees the photo of her destination and gets Directions to find it.

Competitors

	Column 1	Column 2	Column 3	Column 4	Column 5
1	**Feature**	**Google Maps**	**HS-Aalen PDF**	**MazeMap / Pointr**	**CampusNow (Project App)**
2	Indoor Floor Plans	Limited / None	Static (High Detail)	Interactive (Digital)	Interactive / Searchable
3	Blue Dot Nav	Outdoor only	None	Precise (Wi-Fi/BLE)	Hybrid
4	Timetable Sync	No	No	Rare / Integration only	Import / API
5	Guest / Public Mode	Yes	Yes	Limited (usually App)	Full (No Account Needed)
6	Room Photos	None / Limited	No	Sometimes	Yes / Door Photos
7	Event Layer	None	None	No	Yes

- **Direct Competitors:** MazeMap, Pointr (these are professional indoor navigation systems).
- **Indirect Competitors:** Google Maps (for outdoor), Apple Maps, or even the current HS-Aalen website/PDF.
- **Niche Apps:** Eventbrite or Meetup (to see how they handle the "Event Info" part of your brief).

MazeMap

Turn-by-Turn Indoor Navigation | Seaml...
YouTube

Pointr

Introducing Pointr Maps
YouTube

CampusNow Technologie Stack

- **1. Frontend: Flutter**
  - **Framework:** Flutter SDK
  - **Key-Libraries:**
    - **Maps:** Panorama Version: 0.4.1 (Panorama Viewer).

- **2. Backend:**
  - **Language:** Python 3.12
  - **Database:** PostgreSQL 16
    - ***Structure:*** A collection of Buildings, Rooms, and Events.
  - **API:** FastAPI (REST API + OpenAPI/Swagger): https://streetview.8xc.de/docs

- **3. Integrationen und Services:**
  - iCalendar-Parsing mit icalendar
  - Bildverarbeitung mit Pillow

- **4.Infrastruktur:**
  - Docker
  - Nginx (Reverse Proxy, statische Dateiauslieferung)

360° Indoor Navigation - H...
8xc

Key Scenarios

Scenario 1: The "Class-to-Class" Sprint (Primary)
User: Leon (The Anxious Freshman)
Goal: Find the specific room for his next lecture starting in 5 minutes.

1. **Entry Point:** Leon opens the app to the **Dashboard**.
2. **Information Retrieval:** The Dashboard displays "Next: *Embedded Systems* @ Room G2 0.21" (pulled from iCal).
3. **Action:** Leon taps the **"Navigate"** button next to the class name.
4. **Map Transition:** The app switches to the **Map View**, centers on Building G2, and draws a polyline from Leon's GPS "Blue Dot" to the entrance.
5. **Floor Change:** As Leon enters the building, he uses the **Floor Switcher** (UI overlay) to select "Floor 0."
6. **Success:** The room "0.21" highlights in green on the map.

Scenario 2: The "External Meetup" (Guest)
User: Martha (The Concerned Visitor)
Goal: Meet her son at the Cafeteria (Mensa) without having a student account.

1. **Entry Point:** Martha clicks a opens the "CampusNow" location G2.
2. **Onboarding:** The app has only Guest mode.
3. **Visual Confirmation:** The app displays the **Building Detail Panel** for the Mensa, featuring a high-quality photo of the front door.
4. **Task:** Martha taps **"my location"** and inputs here location
5. **Navigation:** She uses the static map to walk toward the visual landmark (the building photo) she saw in the app.

Scenario 3: The "Schedule Onboarding" (First-Time Setup)
User: Leon (New User)
Goal: Connect his personal university schedule to the interactive map.

1. **Action:** Leon goes to the **Settings/Profile** tab.
2. **Instruction:** The app provides a 1-sentence guide: *"Copy your StarPlan iCal link from the HS-Aalen portal."*
3. **Input:** Leon pastes the URL into a text field and hits **"Sync."**
4. **Validation:** The app parses the .ics file, identifies room strings (e.g., "H1.02"), and cross-references them with the database.
5. **Feedback:** The app shows a success message: *"classes synced to your map."*

Product Brief: CampusNow

Der CampusNow ist eine mobile Anwendung, die Studierenden und Hochschulangehörigen eine interaktive, visuelle Orientierungshilfe bietet. Ziel ist es, die Hürden beim Finden von Räumen zu senken und das Campus-Erleben durch integrierte Features wie Stundenpläne und Event-Infos zu zentralisieren.

Problem Statement

Die aktuelle Orientierung am Campus ist für Nutzer unbefriedigend. Hauptprobleme sind:

Mangelnde Orientierung: Insbesondere Erstsemester finden sich auf dem Gelände nicht zurecht.

Statische Ressourcen: Vorhandene Lagepläne sind nicht interaktiv und erschweren die aktive Navigation.

Fragmentierung: Die Suche nach spezifischen Raumplänen ist mühsam und zeitaufwendig.

Importance of Solving the Problem Now

Die ersten Wochen eines Semesters sind entscheidend für die Bindung der Studierenden. Wenn die grundlegende Infrastruktur (Raumfindung) Stress verursacht, sinkt die Zufriedenheit sofort. In einer hybriden Lernwelt erwarten Nutzer zudem moderne, digitale Lösungen – ein statischer PDF-Plan ist nicht mehr zeitgemäß und wirkt abschreckend auf potenzielle neue Studierende.

Data Points and Potential Churn

Hohe Drop-out Quote bei Erstsemestern: Orientierungslosigkeit führt zu Frust und dem Verpassen von Vorlesungen.

Nutzungsabbruch: Ohne interaktive Karte greifen Nutzer auf ineffiziente Workarounds zurück oder meiden optionale Campus-Events komplett, da sie die Veranstaltungsorte nicht finden.

Support-Overload: Manuelle Anfragen bei der Verwaltung und den Fachbereichen binden Personalressourcen, die durch eine Self-Service-App entlastet würden.

Opportunity

Durch die Lösung dieser Probleme steigert die Hochschule ihren ROI durch:

Effizienzgewinn: Weniger Zeitverlust bei der Raumsuche bedeutet mehr Zeit für Lehre und Forschung.

Attraktivität: Ein moderner digitaler Campus ist ein starkes Marketingargument für die Gewinnung neuer Studierender.

Zentralisierung: Die App dient als Hub für Events und Profile, was das Engagement innerhalb der Hochschulgemeinschaft erhöht.

Risks

Datenpflege: Raumpläne und Stundenpläne müssen ständig aktuell gehalten werden (Schnittstellenproblematik).

Akzeptanz: Wenn die App nicht intuitiv genug ist, verbleiben Nutzer bei alten Gewohnheiten.

Technische Hürden: Integration von „Streetview"-ähnlichen Funktionen ist entwicklungsintensiv.

Target Audience

Primär: Erstsemester (Freshmen) und Austauschstudierende.

Sekundär: Bestehende Studierende sowie Vereine und Hochschulorganisationen (für Event-Sichtbarkeit).

Out of Scope: Externe Besucher ohne Bezug zur Hochschule (vorerst).

Definition of Done

Das Projekt gilt als abgeschlossen, wenn:

Die Kernfeatures (Interaktiver Lageplan, Stundenplan-Integration, Event-Bereich) voll funktionsfähig sind.

Die App für iOS/Android verfügbar ist.

Bilder des Campus und Profile für alle relevanten Gebäude hinterlegt sind.

Eine automatisierte Synchronisation mit den Raumbelegungsplänen besteht.

Success Metrics / OKRs

Ziel (Objective)

Kennzahl (Key Result)

Exzellente Onboarding-Erfahrung

90% der Erstsemester nutzen die App in der ersten Woche.

Hohe Usability

Reduktion der Suchanfragen zur Raumfindung im Sekretariat um 40%.

Aktives Campus-Leben

Steigerung der Klickzahlen auf gelistete Campus-Events um 25%.